

Source Code

PDA (Pushdown Automaton) for $L = \{ a^n b^n \mid n \geq 0 \}$ over $\{a, b\}$

Z0 = "\$"

A = "A"

```
def pda(string: str) -> bool:
```

```
    stack = [Z0]
```

```
    state = "q0"
```

```
    print(f'{"Symbol":<8} {"State":<8} Stack (top -> bottom)')
```

```
    print(f'{"-":<8} {"q0":<8} {"".join(reversed(stack))}')
```

```
    for ch in string:
```

```
        top = stack[-1]
```

```
        if state == "q0":
```

```
            if ch == "a":
```

```
                stack.append(A)
```

```
            elif ch == "b":
```

```
                if top != A:
```

```
                    return False
```

```
                stack.pop()
```

```
                state = "q1"
```

```
            else:
```

```
                print(f'Invalid symbol: {ch!r}')
```

```
                return False
```

```
        elif state == "q1":
```

```
            if ch == "b":
```

```
                if top != A:
```

```
                    return False
```

```
                stack.pop()
```

```
else:
```

```
    return False
```

```
    print(f'{ch:<8} {state:<8} {".join(reversed(stack))}")
```

```
    return len(stack) == 1 and stack[0] == Z0
```

```
s = input("Enter a string over {a, b}: ")
```

```
if pda(s):
```

```
    print("\nAccepted")
```

```
else:
```

```
    print("\nRejected")
```

Output

```
Enter a string over {a, b}: aaabbb
Symbol  State   Stack (top -> bottom)
-       q0      $
a       q0      A$
a       q0      AA$
a       q0      AAA$
b       q1      AA$
b       q1      A$
b       q1      $
```

Accepted

○ (base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works>

```
Enter a string over {a, b}: abababa
Symbol  State   Stack (top -> bottom)
-       q0      $
a       q0      A$
b       q1      $
```

Rejected

○ (base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works>

Conclusion

The program successfully implements a PDA that accepts strings of the form $a^n b^n$ with equal numbers of as followed by bs. It rejects strings with unequal numbers or an incorrect ordering of symbols.